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Question 1 — Decision Analysis with Market Research

A firm must choose between two actions (A1 and A2) under two possible states of the world (S1 and S2). The prior probability that the market is favourable (S1) is 0.40; that it is unfavourable (S2) is 0.60. A research firm can provide an imperfect prediction for a fee of $100.

Part (a) — Expected Monetary Value (EMV) Without Research

The EMV is a weighted average of all possible payoffs, where the weights are the prior probabilities of each state occurring.

EMV Formula: $EMV(A_i) = \Sigma_j [\text{ Payoff } (A_i, S_j) \times P(S_j)]$

$$EMV(A1) = (\$400 \times 0.40) + (-\$100 \times 0.60) = \$160 - \$60 = \$100$$
$$EMV(A2) = (\$0 \times 0.40) + (\$100 \times 0.60) = \$0 + \$60 = \$60$$

ACTION / STATE	S1 (FAVOURABLE) P=0.4	S2 (UNFAVOURABLE) P=0.6	EMV ($)
A1 – INVEST	$400	−$100	**$100**
A2 – STATUS QUO	$0	$100	$60

Choose A1, since it has the highest EMV at $100.

Part (b) — Expected Value of Perfect Information (EVPI)

EVPI measures the maximum amount we should ever pay for any information. It represents the gain from having a perfect oracle that reveals which state will occur before we decide. EVPI is an upper bound: no real-world information source can be worth more.

Step 1 — Expected Value WITH Perfect Information (EVwPI):

If we knew S1 would occur → choose A1 (payoff = $400). If we knew S2 would occur → choose A2 (payoff = $100).

$$EVwPI = (\$400 \times 0.40) + (\$100 \times 0.60) = \$160 + \$60 = \$220$$

Step 2 — EVPI:

$$EVPI = EVwPI - \text{Best EMV without info} = \$220 - \$100 = \$120$$

Answer: $120 — We would never pay more than $120 for any market information

Part (c) — Posterior Probabilities via Bayes' Theorem

The research is imperfect: it correctly identifies a favourable market 60% of the time (P(predicts S1 | true S1) = 0.60) and correctly identifies an unfavourable market 80% of the time (P(predicts S2 | true S2) = 0.80). After observing the prediction, we update our prior probabilities using Bayes' Theorem.

Step 1 — Marginal probabilities of each prediction signal:

$$P(I_1 \text{ — predicts S1}) = (0.60 \times 0.40) + (0.20 \times 0.60) = 0.24 + 0.12 = 0.36$$
$$P(I_2 \text{ — predicts S2}) = (0.40 \times 0.40) + (0.80 \times 0.60) = 0.16 + 0.48 = 0.64$$

Signal I_1 — Research predicts S1 (Favourable market):

Prior State	P(State)	$P(I_1 \mid State)$	Joint P	Posterior $P(State \mid I_1)$
S1	0.40	0.60	0.24	**0.667**
S2	0.60	0.20	0.12	**0.333**
Total	**1.00**		**$P(I_1)$ =0.36**	**1.000**

Signal I_2 — Research predicts S2 (Unfavourable market):

Prior State	P(State)	$P(I_2 \mid State)$	Joint P	Posterior $P(State \mid I_2)$
S1	0.40	0.40	0.16	**0.250**
S2	0.60	0.80	0.48	**0.750**
Total	**1.00**		**$P(I_2)$ =0.64**	**1.000**

Posterior — Given I_1: $P(S1|I_1) = 0.667$ | $P(S2|I_1) = 0.333$
Posterior — Given I_2: $P(S1|I_2) = 0.250$ | $P(S2|I_2) = 0.750$

Part (d) — Decision Tree Analysis: Should We Conduct Research?
Using the posterior probabilities, we compute the optimal EMV for each research signal, then weight those by the signal probabilities to find the Expected Value with Research Information (EVwRI).

EMV calculation given each research signal:
Given I_1 (research predicts S1): Choose A1 → EMV = 0.667×\$400 + 0.333× (−\$100) = \$266.80 − \$33.30 = \$233.33
Choose A2 → EMV = 0.667×\$0 + 0.333×\$100 = \$33.33 (A1 is better ✓)

Given I_2 (research predicts S2): Choose A1 → EMV = 0.250×\$400 + 0.750× (−\$100) = \$100 − \$75 = \$25
Choose A2 → EMV = 0.250×\$0 + 0.750×\$100 = \$75 (A2 is better ✓)

Signal	Best Action	P(Signal)	Posterior Best EMV	Weighted EMV	Net (−\$100)
I_1 (pred S1)	A1	0.36	0.667×400 + 0.333× (−100) = \$233.33	0.36 × 233.33 = \$84.00	
I_2 (pred S2)	A2	0.64	0.25×0 + 0.75×100 = \$75.00	0.64 × 75 = \$48.00	
Totals		**1.00**	**EVwRI = \$132.00**	**\$132.00**	**\$32.00**

EVSI = EVwRI − Best EMV without research = \$132 − \$100 = \$32
Net benefit of research = EVSI − Research cost = \$32 − \$100 = −\$68
Final Decision (Part d): Do NOT conduct the research. Choose A1 directly. Net EMV = \$100 > (\$132 − \$100) = \$32

Question 2 — Markov Chain: The Coin Matching Game
Two players (A and B) each start with 2 coins (total = 4 coins). Each trial: both flip one coin simultaneously. If both show the same face (HH or TT), A wins 1 coin from B. Otherwise B wins 1 coin from A. The game ends when one player has all 4 coins. Y_n = number of coins held by Player A after the nth trial.
Part (a) — Is Y_n a Markov Chain? States and Transition Matrix
Yes, Y_n is a Markov chain. The Markov property holds because the outcome of the next trial — and therefore where Y_{n+1} lands — depends only on Y_n (the current number of coins A holds), not on the history of how A accumulated them. Knowing the present state is sufficient to predict the future.

States: S = {0, 1, 2, 3, 4} (number of coins held by Player A)
States 0 and 4 are absorbing states — once reached, the game ends and the chain stay there permanently.
Transition probabilities:
For any non-absorbing state k ∈ {1, 2, 3}:
P (A wins a coin) = P(HH) + P(TT) = 1/4 + 1/4 = 1/2
P (B wins a coin) = P(HT) + P(TH) = 1/4 + 1/4 = 1/2

From \ To	0	1	2	3	4
State 0 (absorb)	1	0	0	0	**0 (A is ruined — absorbing state)**
State 1	1/2	0	1/2	0	**0**
State 2	0	1/2	0	1/2	**0**
State 3	0	0	1/2	0	**1/2**
State 4 (absorb)	0	0	0	0	**1 (A has won — absorbing state)**

Markov Chain: Yes — 5 states {0,1,2,3,4}; absorbing states at 0 and 4; transition prob = 1/2 from all interior states
Part (b) — Limiting Behaviour as n → ∞
Does lim $P^{(n)}$ exist? Yes. Because the game must end in finite expected time (it is a finite absorbing Markov chain), the n-step transition matrix $P^{(n)}$ converges to a limiting matrix $P^{(\infty)}$.

Do the rows of $P^{(\infty)}$ become identical? No. This chain is NOT ergodic (not irreducible). It has two absorbing states, making it an absorbing chain. The row to which the system converges depends entirely on the starting state. For a fair random walk on {0,1,...,N} with N=4, the probability of absorption into state 4 (player A wins) starting from state k is exactly k/N = k/4.

Therefore, the limiting matrix is:

Start State	0	1	2	3	4 (+ note)
0	1	0	0	0	1 (already ruined — will stay ruined)
1	3/4	0	0	0	1/4 — only 25% chance of winning from 1 coin
2	1/2	0	0	0	1/2 — fair game from mid-point
3	1/4	0	0	0	3/4 — 75% chance of winning from 3 coins
4	0	0	0	0	1 (already won)

A player starting with only 1 coin has just a 25% chance of winning. A player starting with 3 coins has a 75% chance. Starting equally (2 coins each) gives a fair 50-50 game. Because rows are different, there are no unconditional (stationary) limiting probabilities the chain cannot have a unique stationary distribution as it is not irreducible.
Limiting Behaviour: P (∞) exists but rows are NOT identical. No stationary distribution exists — this is an absorbing (non-ergodic) chain.
Question 3 — Monte Carlo Simulation: Stock Price (30 Days)

A stock currently trades at $48.00. Each day its price direction is sampled from a multinomial distribution, then the magnitude of change is sampled from a separate distribution conditional on the direction. We simulate 30 trading days.

Simulation Methodology

Step 1 — Assign random number ranges for Direction:

Direction	Probability	Cumulative	RN Range (00–99)
Increase	0.44	0.44	00 – 43
Same	0.23	0.67	44 – 66
Decrease	0.33	1.00	67 – 99

Step 2 — Sample the change amount conditional on direction using separate distributions for Increase/Decrease (as provided in the problem probability tables).

Step 3 — For each of 30 days: draw a direction RN, draw a change-amount RN (if direction ≠ Same), and update the closing price.

30-Day Simulation Results (Starting Price = $48.000)

Day	Direction	Change ($)	Closing Price ($)
0	—	—	48.000
1	Same	0.000	48.000
2	Increase	+0.125	48.125
3	Increase	+0.500	48.625
4	Decrease	−0.750	47.875
5	Increase	+0.250	48.125
6	Increase	+0.125	48.250
7	Same	0.000	48.250
8	Increase	+0.125	48.375
9	Same	0.000	48.375
10	Same	0.000	48.375
11	Increase	+0.375	48.750
12	Decrease	−0.125	48.625
13	Decrease	−0.625	48.000
14	Increase	+0.125	48.125
15	Decrease	−0.375	47.750
16	Increase	+0.125	47.875
17	Decrease	−0.500	47.375
18	Decrease	−0.625	46.750
19	Same	0.000	46.750
20	Decrease	−0.375	46.375
21	Same	0.000	46.375
22	Decrease	−0.500	45.875
23	Decrease	−0.500	45.375
24	Decrease	−0.125	45.250
25	Increase	+0.125	45.375
26	Increase	+0.125	45.500
27	Increase	+0.125	45.625
28	Same	0.000	45.625
29	Increase	+0.125	45.750
30	Increase	+0.125	**45.875**

Simulated Closing Price after 30 days: $45.875 (Net change: −$2.125)

In a manual simulation, the specific closing price depends on the random numbers drawn from the table provided. The methodology above is the correct procedure; the exact price will vary by the random number sequence used.

Question 4 — Queueing Theory: M/M/1/K Barbershop

A barbershop has 1 barber, a waiting room for 1 customer, and turns away arrivals when both the chair and the waiting room are occupied. This is an M/M/1/K queue with K=2 (maximum capacity = 2 customers in the system). Arrivals follow a Poisson process (rate λ), service times are exponentially distributed (rate μ).

Part (a) — Limiting (Steady-State) Probabilities: General Formulas

Using birth-death balance equations, for each state n we equate the rate of leaving state n to the rate of entering it. For a finite-capacity M/M/1/K queue with $\rho = \lambda/\mu$:

Balance equations (flow rate out = flow rate in for each state):

State 0: $\lambda \pi_0 = \mu \pi_1 \quad \rightarrow \pi_1 = \rho \pi_0$

State 1: $(\lambda+\mu) \pi_1 = \lambda \pi_0 + \mu \pi_2 \quad \rightarrow \pi_2 = \rho^2 \pi_0$

State 2: $\mu \pi_2 = \lambda \pi_1$ (boundary)

Normalisation condition (all probabilities must sum to 1):

$$\pi_0 + \pi_1 + \pi_2 = 1$$

$$\pi_0 (1 + \rho + \rho^2) = 1$$

$$\boldsymbol{\pi_0 = 1 / (1 + \rho + \rho^2) \quad | \quad \pi_1 = \rho / (1 + \rho + \rho^2) \quad | \quad \pi_2 = \rho^2 / (1 + \rho + \rho^2)}$$

where $\rho = \lambda/\mu$ (traffic intensity)

Part (b) — Numerical Results: λ = 3 customers/hr, μ = 4 customers/hr

$\rho = \lambda/\mu = 3/4 = 0.75$

Denominator $= 1 + 0.75 + 0.75^2 = 1 + 0.75 + 0.5625 = 2.3125 = 37/16$

Metric	Formula	Value (λ=3, μ=4, ρ=0.75)
π_0 (shop empty)	$1 / (1 + \rho + \rho^2) = 1 / 2.3125$	16/37 ≈ 0.4324 (43.24%)
π_1 (one customer)	$\rho \cdot \pi_0 = 0.75 \times 0.4324$	12/37 ≈ 0.3243 (32.43%)
π_2 (shop full)	$\rho^2 \cdot \pi_0 = 0.5625 \times 0.4324$	9/37 ≈ 0.2432 (24.32%)
L (avg customers in system)	$0 \cdot \pi_0 + 1 \cdot \pi_1 + 2 \cdot \pi_2 = 12/37 + 18/37$	30/37 ≈ 0.8108 customers
% who enter shop	$1 - \pi_2$	28/37 ≈ 75.68%
% who balk (shop full)	π_2	9/37 ≈ 24.32%

Average customers in shop (L): 30/37 ≈ 0.811 customers

Proportion of customers who enter: 28/37 ≈ 75.68% (24.32% are turned away — shop is full)

Question 5 — Dynamic Programming: Critical Path Analysis

A project consists of activities connected in a network. We use backwards induction (dynamic programming) to find the longest path from Start to Finish, which determines the minimum project duration. The critical path is the sequence of activities that must all complete on time — any delay in this path delays the entire project.

Network Summary

Activity	Duration (weeks)	Predecessors	Successors
Start	0	—	A, B
A	5	Start	C, D
B	3	Start	E
C	4	A	F, G
D	2	A	H
E	3	B	H, I
F	1	C	J
G	4	C	K
H	6	D, E	K, L
I	2	E	L
J	5	F	Finish

K	4	G, H	Finish
L	7	H, I	Finish
Finish	0	J, K, L	—

Stages and States

We work backwards from Finish, processing activities layer by layer. The state at each stage is the activity node currently being considered.

Stage (n)	Nodes / States at this Stage	Distance from Finish
n = 1	J, K, L	Directly precede Finish
n = 2	F, G, H, I	Two steps from Finish
n = 3	C, D, E	Three steps from Finish
n = 4	A, B	Four steps from Finish

Recursive Function

The recursive (Bellman) equation for the longest path is:

f*(s) = Duration(s) + max { f*(t) : t is a direct successor of s }

Base case: f*(Finish) = 0

The longest path from node s to Finish equals the time to complete activity s itself, plus the longest remaining path from the best successor. We pick the successor that maximises this total.

DP Tables — Backward Induction

Stage n = 1 (nodes: J, K, L — all connect directly to Finish)

State (s)	Duration (weeks)	→ Finish f*(Finish)=0	f*(s) Longest path	Optimal next
J	5	5 + 0 = 5	5	Finish
K	4	4 + 0 = 4	4	Finish
L	7	7 + 0 = 7	7	Finish

Stage n = 2 (nodes: F, G, H, I)

State (s)	Duration (weeks)	Option 1	Option 2	f*(s)	Optimal next
F	1	→J: 1+5 = 6	—	**6**	J
G	4	→K: 4+4 = 8	—	**8**	K
H	6	→K: 6+4 = 10	→L: 6+7 = 13	**13**	L
I	2	→L: 2+7 = 9	—	**9**	L

Stage n = 3 (nodes: C, D, E)

State (s)	Duration (weeks)	Option 1	Option 2	f*(s)	Optimal next
C	4	→F: 4+6 = 10	→G: 4+8 = 12 ✓	**12**	G
D	2	→H: 2+13 = 15	—	**15**	H
E	3	→H: 3+13 = 16 ✓	→I: 3+9 = 12	**16**	H

Stage n = 4 (nodes: A, B)

State (s)	Duration (weeks)	Option 1	Option 2	f*(s)	Optimal next
A	5	→C: 5+12 = 17	→D: 5+15 = 20 ✓	**20**	D
B	3	→E: 3+16 = 19	—	**19**	E

Critical Path — Forward Trace

Starting from Start, follow the optimal successor at each stage:

Activity	Duration	Reason for inclusion
Start	0	Origin node
A	5	f*(A) = 20 > f*(B) = 19 → choose A from Start
D	2	f*(D)+5 = 20 > f*(C)+5 = 17 → choose D from A
H	6	Only successor of D (and maximises f* at n=2)
L	7	f*(L) = 7 > f*(K) = 4 → choose L from H
Finish	0	Terminal node

Total Duration = 0 + 5 + 2 + 6 + 7 + 0 = 20 weeks

Critical Path: Start → A → D → H → L → Finish

Minimum Project Duration: 20 weeks

Verification: Next longest path is Start→ B → E → H → L→ Finish = 3+3+6+7 = 19 weeks. Only one critical path exists. Any delay in A, D, H, or L directly delays the entire project.